UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Legendre's and Cauchy's Linear Differential Equations

Complete study pack for class 011, Legendre (the deck's SESSION 9): linear equations with variable coefficients $k_0(ax+b)^n y^{(n)} + \cdots + k_n y = X$ reduced to constant coefficients by $ax+b = e^z$, with $(ax+b)\frac{dy}{dx} = aDy$, $(ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y$; Cauchy's equation as the case $a=1$, $b=0$ ($x=e^z$); the slide example, the three lecture examples, all five class-work problems and all six distinct ESA questions (6–7 marks each) solved in full and checked with sympy

Compiled from the PESU Unit 2 deck “Legendre's differential equation” (bucket 011, 11 slides, title slide “SESSION: 9”), its two AV-summary transcripts (T011L, 21.9 min; T011Lb “Continued”, 23 min), the Unit 1 & 2 class-work sheet (problems 29–33) and eight past ESA papers. Generated 17 October 2026.

· How to use this pack

Topic 1: what a Legendre or Cauchy equation is (recognise it: degree of the coefficient = order of the derivative). Topic 2: the substitution and the operator results (E2.1–E2.6, F2.1) — the whole method. Topic 3: Legendre problems (slide example, CW P29–P30, five ESA questions). Topic 4: Cauchy problems (lecture examples, CW P31–P33, ESA Jan-24 s3 3d). After the substitution every problem is a constant-coefficient equation in z : CF cases are in the class 002 pack, the PI rules in classes 003 (e^{kz}, incl. failure), 004 ($\sin$, $\cos$), 005 (polynomials in z), 006 ($e^{kz}V$, zV), 009 ($\frac{1}{D-m}X = e^{mz}\int Xe^{-mz}dz$) and variation of parameters in the class 011 VoP pack. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011L·8	Unit 2 deck 011 “Legendre's differential equation” (SESSION 9), slide 8 (rendered .pptx)
T011L 00:05:20	AV-summary transcript “Legendre differential equation” (21.9 min, listed first), segment start
T011Lb 00:12:26	AV-summary transcript “Legendre differential equations_Continued” (23 min, second), segment start
CW U2 P30	Unit 1 & 2 class-work sheet, Unit 2 problem 30
ESA Jul-23 3a	Past paper: UE22MA141A July 2023, question 3.a (Jan-24 s1...s4 = UE23MA141A Jan 2024 sets 1–4)

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4. 4 · Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 Linear equations with variable coefficients: Legendre's and Cauchy's forms

So far the coefficients were constants. This session treats two forms with variable coefficients that a substitution turns into constant-coefficient equations. Legendre's equation has the coefficient $(ax+b)^k$ in front of the k-th derivative; Cauchy's (Cauchy–Euler) equation has x^k, and is Legendre's with $a=1$, $b=0$. The test: in every term the degree of the coefficient equals the order of the derivative.

Key terms and definitions

Legendre's linear differential equation. An equation of the form $k_0(ax+b)^n\frac{d^ny}{dx^n}+k_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+k_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$, where the k_i are constants ($k_0\neq0$) and X is a function of x; it is of order n. S011L·4 · S011L·5 · T011L 00:01:38 · T011L 00:02:05

Cauchy's linear differential equation. An equation of the form $k_0x^n\frac{d^ny}{dx^n}+k_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+k_2x^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$, X a function of x: Legendre's equation with $a=1$, $b=0$ (also called the Cauchy–Euler or Euler equation). S011L·4 · T011L 00:02:16 · T011L 00:16:51 · T011L 00:17:52

Variable coefficients. The coefficients of the derivatives are functions of the independent variable x (here $(ax+b)^k$ or x^k), not constants. T011L 00:03:10 · S011L·4

> *"We consider two forms of differential equations with variable coefficients which can be reduced to linear differential equations with constant coefficients by suitable substitutions."* S011L·4

Formulas

E1.1 · Legendre's equation of order n ESA ★★★

$$k_0(ax+b)^n\frac{d^ny}{dx^n}+k_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}}$$
$$+k_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$$

gives	The form to recognise.
applies	Power of $(ax+b)$ = order of the derivative in every term; $k_0\neq0$.
note	Slide 011L·5 prints the third derivative as $\frac{d^{n-1}y}{dx^{n-1}}$: it is $\frac{d^{n-2}y}{dx^{n-2}}$.

S011L·4 · S011L·5 · T011L 00:01:38 · T011L 00:04:06

E1.2 · Cauchy's equation of order n ($a = 1$, $b = 0$) ESA ★★★

$$k_0 x^n \frac{d^n y}{dx^n} + k_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + k_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + k_n y = X$$

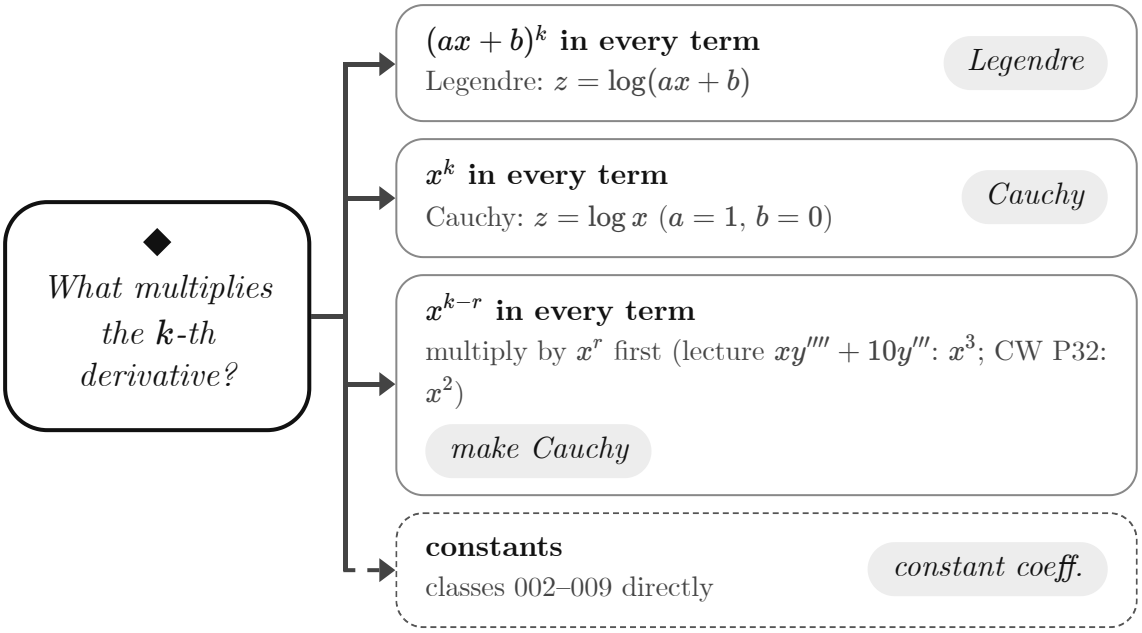
GIVES The special case.

APPLIES Coefficient x^k in front of the k-th derivative.

NOTE Slide 011L·4 also prints $\frac{d^{n-1}y}{dx^{n-1}}$ in the third term; it is $\frac{d^{n-2}y}{dx^{n-2}}$.

S011L·4 · T011L 00:16:51 · T011L 00:17:52 · ESA Jan-24 s3 3d

Flowcharts



Remark. The test is the same for all: the power of the factor equals the order of the derivative.

Figure F1.1. Which form is it? ESA ★★★

S011L·4 · T011L 00:04:06 · T011Lb 00:18:25 · CW U2 P32

Important points and why

1. Recognise the form: in every term the degree of the coefficient equals the order of the derivative ($(ax+b)^2$ with y'', $(ax+b)$ with y', $(ax+b)^0$ with y). ESA ★★★ T011L 00:04:06 · T011L 00:04:46 · T011L 00:15:00 · T011Lb 00:01:31
2. Cauchy's equation is Legendre's with $a = 1$, $b = 0$; the same method works with $x = e^z$. ESA ★★★ S011L·4 · T011L 00:02:16 · T011L 00:17:52 · T011Lb 00:03:24
3. An equation that is not in the form can sometimes be brought to it: $xy'''' + 10y''' = 0$ times x^3 is Cauchy's equation $x^4y'''' + 10x^3y''' = 0$; $y'' + \frac{1}{x}y' = \ldots$ times x^2 (CW P32). ESA ★★ TRANSCRIPT ONLY T011Lb 00:18:25 · T011Lb 00:18:37 · CW U2 P32

4. Cauchy's equation is the syllabus “self-learning component”, but it is asked (ESA Jan-24 s3 3d = s4 3c) and has three CW problems. ESA ★★★ NOT ON CURRENT SLIDES CW U2 P31 · ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

Exam trap. Slides 011L·4–5 print $k_2(ax+b)^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$ (and $k_2x^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$): the derivative is $\frac{d^{n-2}y}{dx^{n-2}}$ — degree equals order. S011L·4 · S011L·5 · T011L 00:04:06

SHORT-NOTE RECAP

Legendre: $k_0(ax+b)^n y^{(n)} + k_1(ax+b)^{n-1}y^{(n-1)} + \cdots + k_n y = X$; Cauchy: the same with x^k ($a=1$, $b=0$). Variable coefficients, but the degree of each coefficient equals the order of its derivative, so a substitution reduces it to constant coefficients. $xy'''' + 10y''' = 0$ becomes Cauchy after multiplying by x^3.

2 The substitution $ax+b=e^z$ and the operator results

Put $z=\log(ax+b)$, so $\frac{dz}{dx}=\frac{a}{ax+b}$, and write $D=\frac{d}{dz}$. Then $(ax+b)\frac{dy}{dx}=aDy$, $(ax+b)^2\frac{d^2y}{dx^2}=a^2D(D-1)y$, $(ax+b)^3\frac{d^3y}{dx^3}=a^3D(D-1)(D-2)y$. The right-hand side is rewritten with $x=\frac{e^z-b}{a}$; the result is a constant-coefficient equation in z, solved by the CF and PI rules of classes 002–009, and the answer is converted back with $e^z=ax+b$. For Cauchy: $x=e^z$, xDy-results with $a=1$.

Key terms and definitions

The substitution. $ax+b=e^z$, i.e. $z=\log(ax+b)$ and $x=\frac{e^z-b}{a}$; for Cauchy $x=e^z$, $z=\log x$. It changes the independent variable from x to z. S011L·5 · T011L 00:05:20 · T011Lb 00:01:49 · T011L 00:18:30

D after the substitution. $D=\frac{d}{dz}$, the differential operator with respect to z (some books write θ); before the substitution D meant $\frac{d}{dx}$. S011L·6 · T011L 00:07:44 · T011Lb 00:02:04

> *"Such equations can be reduced to linear differential equations with constant coefficients by the substitution $ax+b=e^z$ or $z=\log(ax+b)$, $\frac{dz}{dx}=\frac{a}{ax+b}$."* S011L·5

> *"Using these results in Legendre's Differential Equation, we get Linear Differential Equation with constant coefficients which can be solved by the methods discussed before."* S011L·6

> *"Consider Second order Legendre's Linear Differential Equation $(ax+b)^2\frac{d^2y}{dx^2}+k_1(ax+b)\frac{dy}{dx}+k_2y=X$. By the substitution $z=\log(ax+b)$, the above equation reduces to $(a^2D(D-1)+k_1aD+k_2)y=f(z)$. It is a second order linear non homogeneous differential equation with constant coefficients which can be solved."* S011L·7

Formulas

E2.1 · The substitution and $\frac{dz}{dx}$ ESA ★★★

$$ax+b=e^z,\qquad z=\log(ax+b),\qquad x=\frac{e^z-b}{a},\qquad \frac{dz}{dx}=\frac{a}{ax+b}$$

gives	The change of variable.
applies	Every Legendre equation.
note	Cauchy: $x=e^z$, $z=\log x$, $\frac{dz}{dx}=\frac{1}{x}$.

S011L·5 · T011L 00:05:20 · T011L 00:06:11 · T011Lb 00:01:49

E2.2 · First derivative

ESA ★★★

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{a}{ax+b} \quad\Rightarrow\quad (ax+b)\frac{dy}{dx} = aDy, \qquad D = \frac{d}{dz}$$

GIVES The first-order term.

APPLIES Chain rule.

NOTE Cauchy: $x\frac{dy}{dx} = Dy$.

S011L·6 · T011L 00:07:16 · T011L 00:07:44 · T011L 00:18:54

E2.3 · Second derivative

ESA ★★★

$$(ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y = a^2(D^2-D)y$$

GIVES The second-order term.

APPLIES Product rule on $\frac{a}{ax+b}\frac{dy}{dz}$, then the chain rule.

NOTE Cauchy: $x^2\frac{d^2y}{dx^2} = D(D-1)y$. Not a^2D^2y.

S011L·6 · T011L 00:08:53 · T011L 00:11:45 · T011L 00:19:07

E2.4 · Third derivative and the general rule

ESA ★★

$$(ax+b)^3\frac{d^3y}{dx^3} = a^3D(D-1)(D-2)y$$
$$(ax+b)^n\frac{d^ny}{dx^n} = a^nD(D-1)(D-2)\cdots(D-n+1)y$$

GIVES Higher-order terms (n linear factors).

APPLIES Order 3 and 4 equations (lecture examples).

NOTE Slide 011L·6 shows $n = 1, 2, 3$; the general line is added for clarity. Cauchy: $x^4y'''' = D(D-1)(D-2)(D-3)y$.

S011L·6 · T011L 00:12:21 · T011L 00:12:34 · T011Lb 00:20:04

E2.5 · Second-order Legendre equation reduced

ESA ★★★

$$(ax+b)^2y'' + k_1(ax+b)y' + k_2y = X$$
$$\Longrightarrow\ \left(a^2D(D-1) + k_1aD + k_2\right)y = X\left(\tfrac{e^z-b}{a}\right)$$

GIVES A constant-coefficient equation in z.

APPLIES Every ESA question.

NOTE Divide by a^2 if convenient; the AE is $a^2m(m-1) + k_1am + k_2 = 0$.

S011L·7 · T011L 00:14:20 · T011L 00:15:57 · ESA Jul-23 3a

E2.6 · Back to x ESA ★★★

$$e^{mz} = (ax+b)^m, \qquad z = \log(ax+b), \qquad \cos\beta z = \cos(\beta \log(ax+b))$$

GIVES The final answer in x.

APPLIES Last step, always.

NOTE Repeated root m: $(c_1 + c_2 z)e^{mz} = (ax+b)^m[c_1 + c_2 \log(ax+b)]$.

S011L·9 · T011Lb 00:09:11 · T011Lb 00:12:12 · T011Lb 00:17:39

Graphs and diagrams

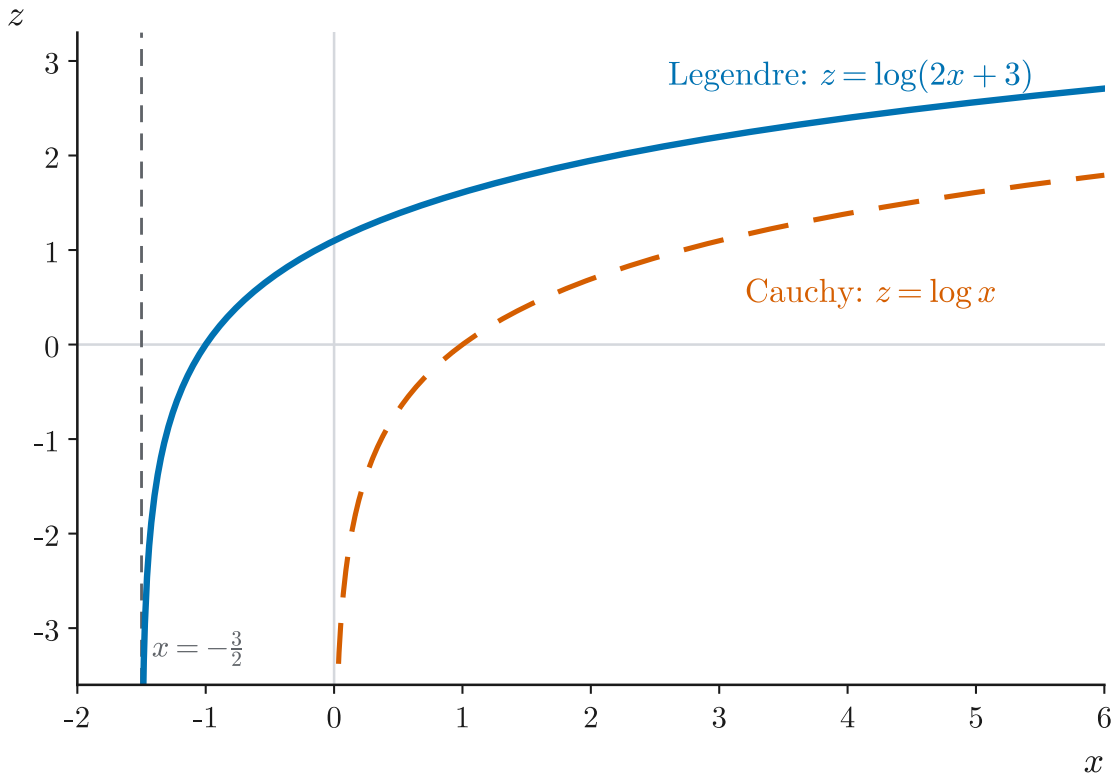


Figure G2.1. The new variable: $z = \log(ax+b)$ and $z = \log x$. ESA ★★

ADDED FOR CLARITY S011L·5 · T011L 00:05:20 · T011L 00:18:30

How to read it. Blue: $z = \log(2x+3)$ (slide example, $a=2$, $b=3$), defined for $x > -\frac{3}{2}$ (grey asymptote). Vermilion dashed: $z = \log x$ (Cauchy), defined for $x > 0$. Both are increasing, so each x has one z and back. Equal steps in z are ever larger steps in x: e^{mz} in z becomes the power $(ax+b)^m$ in x.

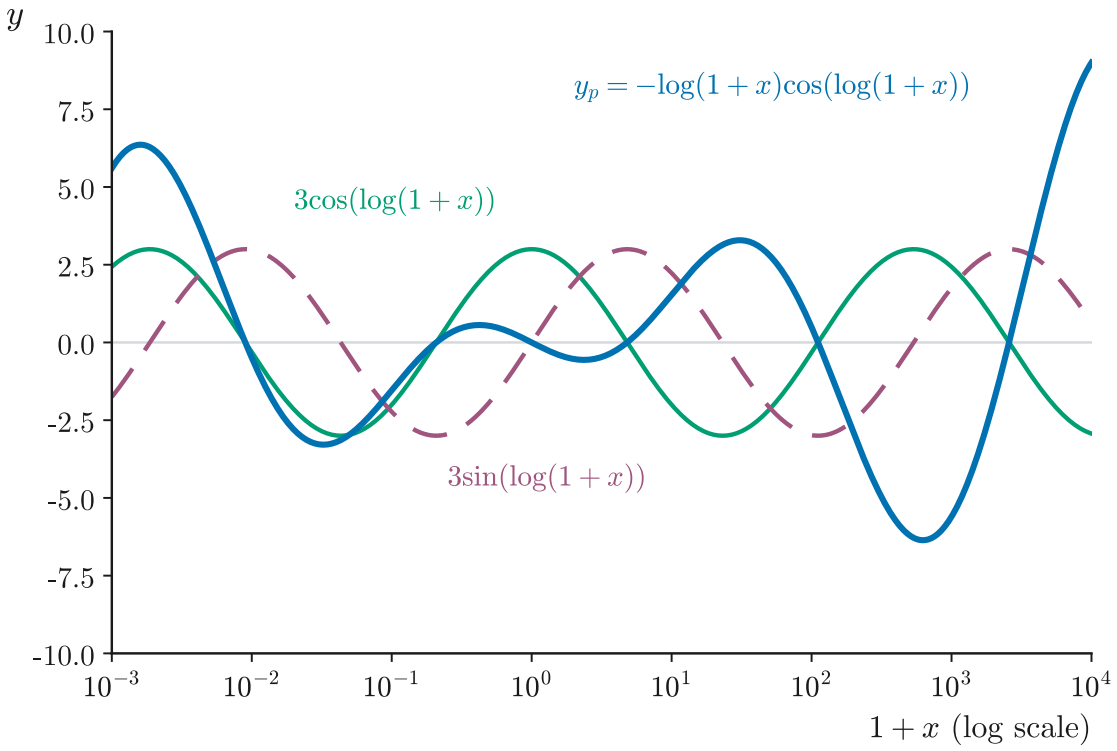
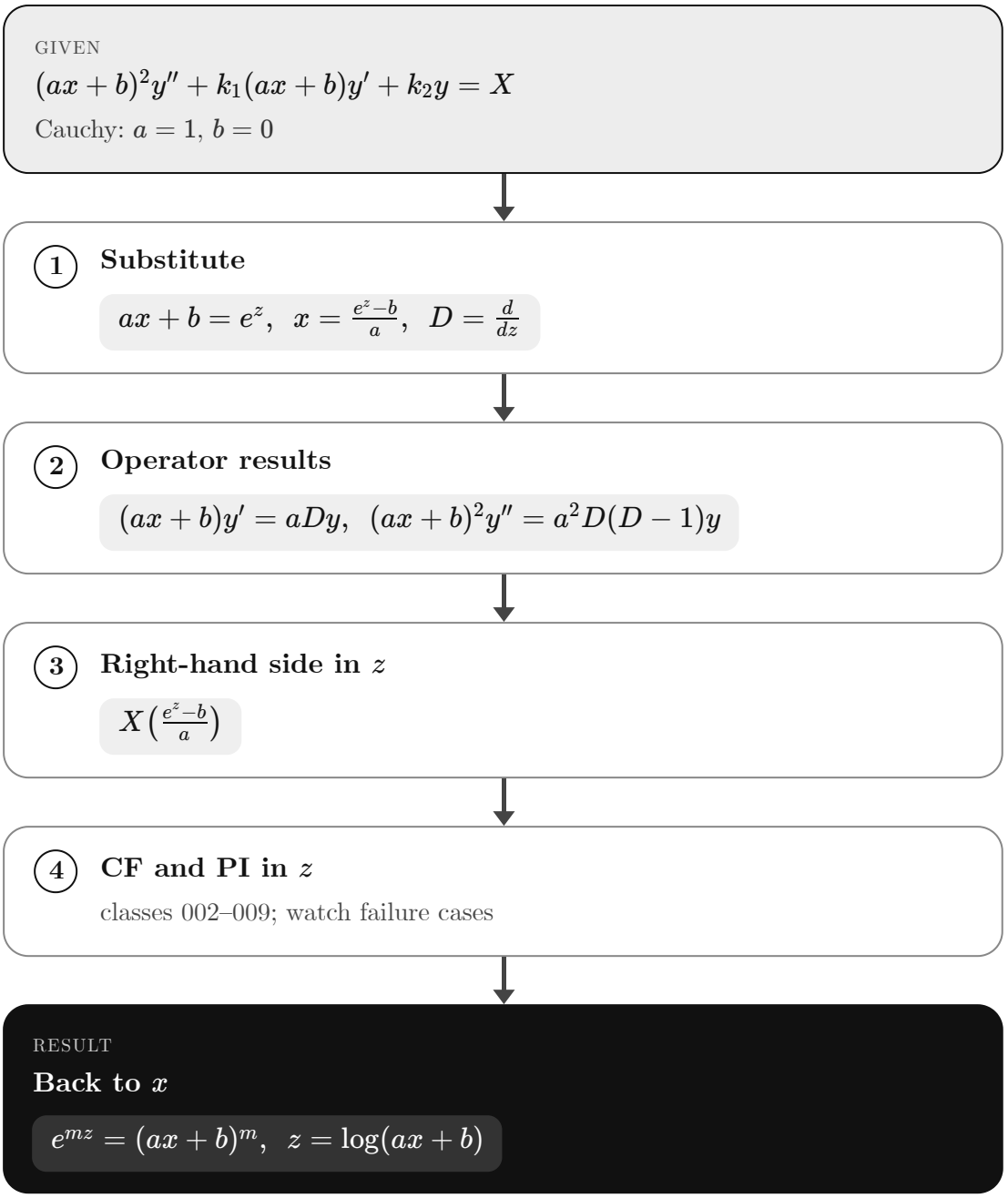


Figure G2.2. Complex roots: $\cos(\log(1+x))$ oscillates ever more slowly. ESA ★★

ADDED FOR CLARITY CW U2 P29 · ESA Dec-23 3b

How to read it. CW P29 and ESA Dec-23 3b have AE roots $\pm i$, so the CF is $c_1\cos(\log(1+x)) + c_2\sin(\log(1+x))$. The horizontal axis is $1+x$ on a log scale, from 10^{-3} to 10^{4}: there the CF terms (green, purple dashed; drawn $\times 3$) are ordinary sinusoids in $z = \log(1+x)$, one period per factor $e^{2\pi} \approx 535$ in $1+x$ — in x the zeros crowd towards $x = -1$ and spread out for large x. Blue: the CW P29 PI $-\log(1+x)\cos(\log(1+x))$, whose amplitude grows like $|\log(1+x)|$ (resonance in z).

Flowcharts



Remark. In steps 2–4 the variable is z: a failure case gives $z\,e^{kz}$.

Exam trap. $a^2D(D-1)$, not a^2D^2 and not $D(D-1)$.

Figure F2.1. Solving a Legendre (or Cauchy) equation ESA ★★★

S011L·5 · S011L·6 · S011L·7 · T011L 00:13:26 · T011Lb 00:12:12

Legendre and Cauchy side by side

	Legendre	Cauchy
Coefficient of $y^{(k)}$	$k_{n-k}(ax+b)^k$	$k_{n-k}x^k$
Substitution	$ax+b=e^z$, $z=\log(ax+b)$	$x=e^z$, $z=\log x$
$\frac{dz}{dx}$	$\frac{a}{ax+b}$	$\frac{1}{x}$
First order	$(ax+b)y'=aDy$	$xy'=Dy$
Second order	$(ax+b)^2y''=a^2D(D-1)y$	$x^2y''=D(D-1)y$
Third order	$a^3D(D-1)(D-2)y$	$D(D-1)(D-2)y$
e^{mz} **in** x	$(ax+b)^m$	x^m
On the slides?	yes (011L·5–10)	form only (011L·4); rest transcript only

S011L·4 · S011L·5 · S011L·6 · T011L 00:18:30 · T011L 00:19:07

Important points and why

1. Four steps (F2.1): substitute $z=\log(ax+b)$ → write the equation and the RHS in z → CF and PI in z → back to x. ESA ★★★ S011L·7 · S011L·8 · T011L 00:13:26 · T011Lb 00:12:12
2. Remember $(ax+b)^2y''=a^2D(D-1)y$, not a^2D^2y: the $-D$ comes from the product rule. ESA ★★★ S011L·6 · T011L 00:11:45 · T011L 00:12:34
3. The factor a matters: $(2x+3)y'=2Dy$ and $(2x+3)^2y''=4D(D-1)y$. ESA ★★★ S011L·8 · T011Lb 00:13:23 · T011Lb 00:13:37
4. The right-hand side must also be written in z (use $x=\frac{e^z-b}{a}$); the final answer must be in x again. ESA ★★★ T011L 00:15:57 · T011L 00:21:15 · T011Lb 00:09:29 · T011Lb 00:17:39
5. After the substitution D means $\frac{d}{dz}$, and a failure case gives $z\,e^{kz}$, not $x\,e^{kx}$. ESA ★★★ T011L 00:07:44 · T011L 00:12:03 · T011Lb 00:11:07

Why does $ax+b=e^z$ remove the variable coefficients?
Since $\frac{dz}{dx}=\frac{a}{ax+b}$, each x-derivative brings a factor $\frac{1}{ax+b}$; the coefficient $(ax+b)^k$ in front of the k-th derivative cancels them exactly, leaving constants times z-derivatives. That is why the degree must equal the order.

ADDED FOR CLARITY T011L 00:06:28 · T011L 00:13:26

Where does the $D-1$ in $a^2D(D-1)$ come from?
$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{a}{ax+b}\frac{dy}{dz}\right)=-\frac{a^2}{(ax+b)^2}\frac{dy}{dz}+\frac{a^2}{(ax+b)^2}\frac{d^2y}{dz^2}$: the first term (derivative of $\frac{a}{ax+b}$) gives $-D$.
T011L 00:09:33 · T011L 00:10:52 · T011L 00:11:45

Why does e^{mz} become a power of $ax+b$?
$e^{mz} = (e^z)^m = (ax+b)^m$. So Legendre and Cauchy solutions are powers (and logs), not exponentials, in x.

S011L·9 · T011Lb 00:15:21

Exam trap. Forgetting the powers of a: $(2x+1)^2y'' = 4D(D-1)y$, not $D(D-1)y$. S011L·8 · ESA Jul-23 3a

Exam trap. Leaving the right-hand side in x (e.g. solving $(D^2-2D-3)y = 6x$ in z) is wrong: write $x = \frac{e^z-3}{2}$ first. S011L·8 · T011L 00:15:57

Exam trap. Answers left in z lose marks: replace e^z by $ax+b$ and z by $\log(ax+b)$. T011Lb 00:09:29 · T011Lb 00:17:39

SHORT-NOTE RECAP

$ax+b = e^z$, $z = \log(ax+b)$, $\frac{dz}{dx} = \frac{a}{ax+b}$, $D = \frac{d}{dz}$: $(ax+b)y' = aDy$, $(ax+b)^2y'' = a^2D(D-1)y$, $(ax+b)^3y''' = a^3D(D-1)(D-2)y$. Write X in z ($x = \frac{e^z-b}{a}$), find CF and PI in z, convert back ($e^{mz} = (ax+b)^m$). Cauchy: $x = e^z$, $xy' = Dy$, $x^2y'' = D(D-1)y$.

3 Legendre problems: the slide example, CW P29–P30 and five ESA questions

Every Legendre question in the papers is second order. A polynomial right-hand side in x becomes a sum of e^{kz} terms and a constant; $\sin(k\log(ax+b))$ becomes $\sin kz$. Four of the five ESA questions hit a PI failure case ($(D-2)^2$, $(D-1)^2$, $D-1$), so the rule $\frac{1}{f(D)}e^{kz} = \frac{z}{f'(k)}e^{kz}$ is needed. CF terms come back as powers $(ax+b)^m$, repeated roots as $(ax+b)^m\log(ax+b)$, complex roots as $\cos(\beta\log(ax+b))$.

Formulas

E3.1 · What the right-hand side becomes

ESA ★★★

$$x = \tfrac{e^z-b}{a}: \quad (ax+b)^k \to e^{kz}, \quad px+q \to \tfrac{p}{a}(e^z-b)+q$$
$$\log(ax+b) \to z, \qquad \sin(k\log(ax+b)) \to \sin kz$$

GIVES The PI type in z.

APPLIES Before the PI.

NOTE A polynomial in x gives a sum of e^{kz} and a constant (Type 1 with $k=0$).

S011L·8 · T011Lb 00:14:09 · CW U2 P30 · ESA Jun-24 3b

E3.2 · PI rules needed in z (classes 003–004)

ESA ★★★

$$\frac{1}{f(D)}e^{kz} = \frac{e^{kz}}{f(k)},\ f(k)\neq 0; \qquad \frac{1}{f(D)}e^{kz} = \frac{z\,e^{kz}}{f'(k)},\ f(k)=0\neq f'(k)$$
$$\frac{1}{(D-k)^2}e^{kz} = \frac{z^2}{2}e^{kz}, \qquad \frac{1}{D^2+\beta^2}\sin\beta z = -\frac{z}{2\beta}\cos\beta z$$

GIVES Every PI in Topic 3.

APPLIES Failure cases in 4 of 6 ESA questions.

NOTE A constant c is ce^{0z}: $\frac{c}{f(0)}$.

ADDED FOR CLARITY T011Lb 00:10:29 · T011Lb 00:16:27 · ESA Jul-23 3a · ESA Jan-24 s2 3c · CW U2 P29

E3.3 · ESA answers (Legendre)

ESA ★★★

$$\text{Jul-23: } y = (2x+1)^2\left[c_1 + c_2\log(2x+1) + \log^2(2x+1)\right]$$
$$\text{Jan-24 s2: } y = (x+2)\left[c_1 + c_2\log(x+2)\right] + \tfrac{3}{2}(x+2)\log^2(x+2) - 2$$

GIVES The two repeated-root ESA answers.

APPLIES W3.5, W3.7.

NOTE Others: W3.3, W3.6, W3.8.

ESA Jul-23 3a · ESA May-24 3a · ESA Jan-24 s2 3c

Graphs and diagrams

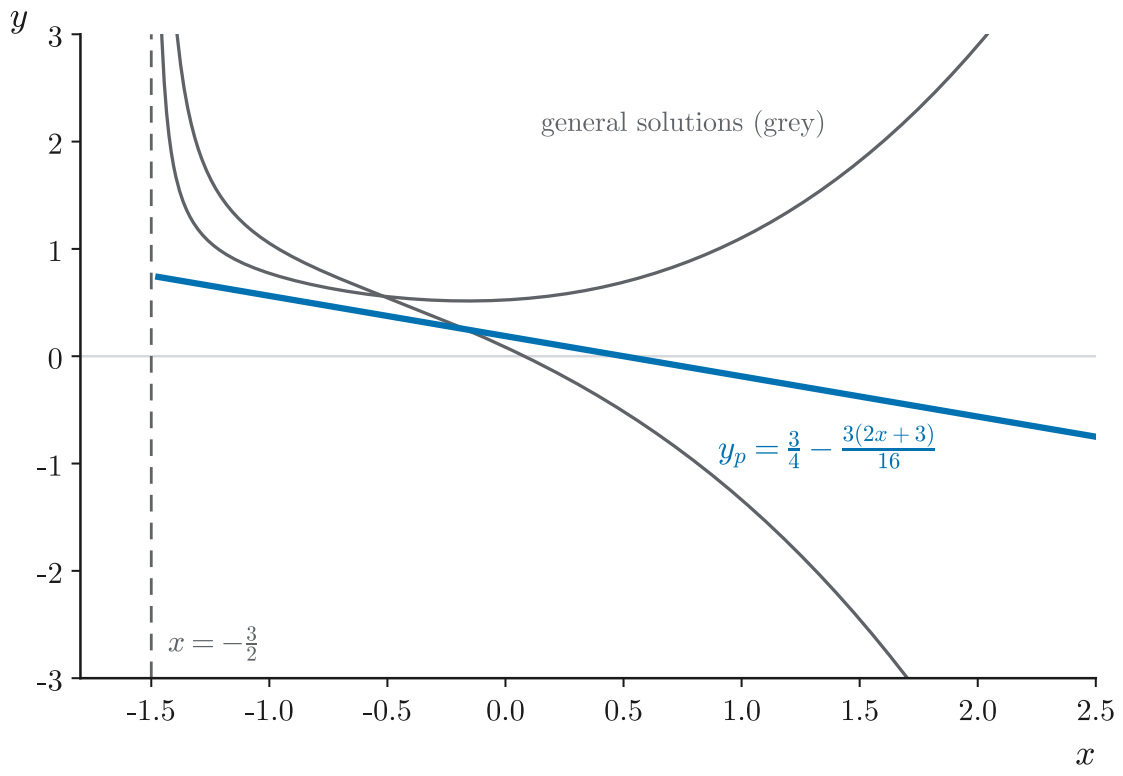
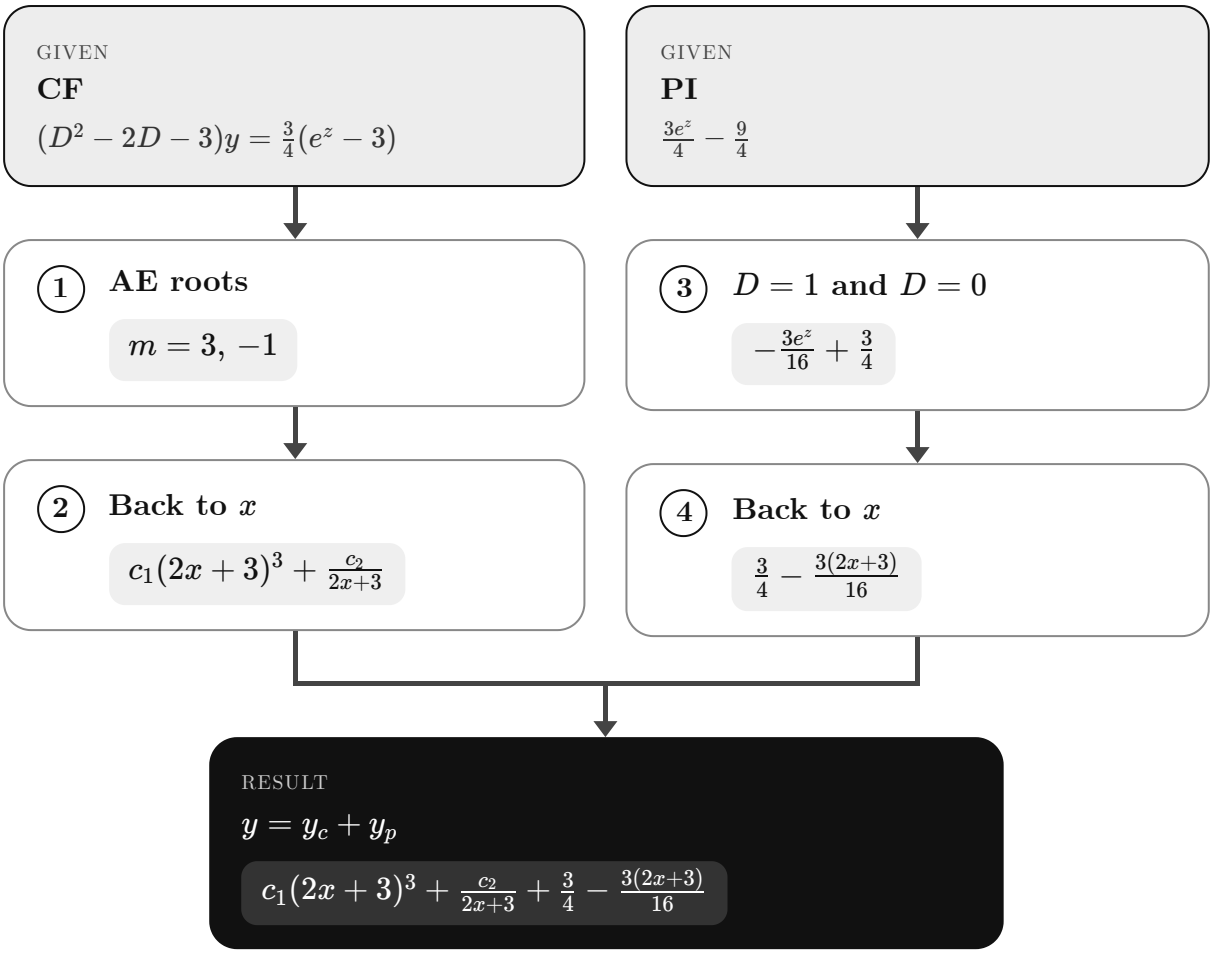


Figure G3.1. Slide example: $y_p = \frac{3}{4} - \frac{3(2x+3)}{16}$ and general solutions. ESA ★★★

ADDED FOR CLARITY S011L·10

How to read it. For $x > -\frac{3}{2}$. Blue: the PI, a straight line $\frac{3}{16} - \frac{3x}{8}$ (a polynomial right-hand side $6x$ gave a polynomial PI). Grey: general solutions with $(c_1, c_2) = (0.01, 0.2)$ and $(-0.01, 0.5)$; the $\frac{c_2}{2x+3}$ term blows up at the asymptote $x = -\frac{3}{2}$ and the $c_1(2x+3)^3$ term dominates for large x.

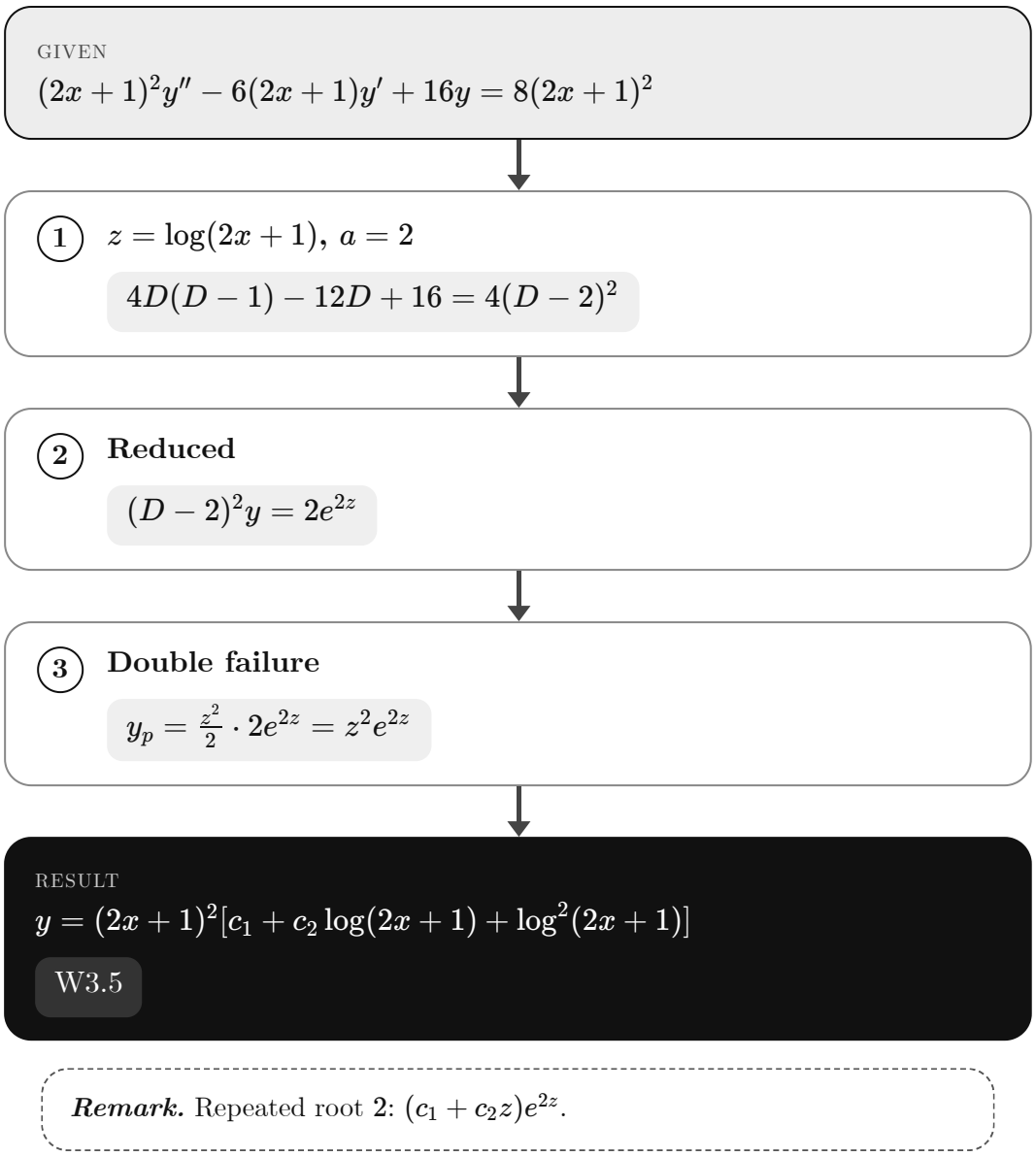
Flowcharts



Remark. Worked in full as W3.1.

Figure F3.1. Slides 011L·8–10: $(2x+3)^2y''-2(2x+3)y'-12y=6x$ ESA ★★★

S011L·8 · S011L·9 · S011L·10



Remark. Repeated root 2: $(c_1 + c_2 z)e^{2z}$.

Figure F3.2. ESA Jul-23 3a = May-24 3a ESA ★★★

ESA Jul-23 3a · ESA May-24 3a

Every problem in this pack, reduced to z

Problem	Equation in z	Roots	PI in x
Slide (W3.1)	$(D^2 - 2D - 3)y = \frac{3}{4}(e^z - 3)$	$3, -1$	$\frac{3}{4} - \frac{3(2x+3)}{16}$
CW P29 (W3.2)	$(D^2 + 1)y = 2\sin z$	$\pm i$	$-\log(1+x)\cos\log(1+x)$
Dec-23 (W3.3)	$(D^2 + 1)y = \sin 2z$	$\pm i$	$-\frac{1}{3}\sin(2\log(1+x))$
CW P30 (W3.4)	$(D^2 - 4)y = \frac{e^{2z}-1}{27}$	± 2	$\frac{(3x+2)^2\log(3x+2)+1}{108}$
Jul-23 (W3.5)	$(D-2)^2y = 2e^{2z}$	$2, 2$	$(2x+1)^2\log^2(2x+1)$
Jan s1 (W3.6)	$4(D^2 - 2D - 3)y = 3e^z + 2$	$3, -1$	$-\frac{3(2x+1)}{16} - \frac{1}{6}$

Problem	Equation in z	Roots	PI in x
Jan s2 (W3.7)	$(D-1)^2y = 3e^z - 2$	$1, 1$	$\frac{3}{2}(x+2)\log^2(x+2) - 2$
Jun-24 (W3.8)	$(4D^2-2D-2)y = 2e^{2z} + 3e^z + 4$	$1, -\frac{1}{2}$	$\frac{(2x-1)^2}{5} + \frac{(2x-1)\log(2x-1)}{2} - 2$
Lecture (W4.1)	$(D^3 - D^2 + 2)y = 10(e^z + e^{-z})$	$-1, 1 \pm i$	$5x + \frac{2\log x}{x}$
CW P31 (W4.3)	$(D^2 - 2D - 3)y = ze^{2z}$	$3, -1$	$-\frac{x^2}{3}(\log x + \frac{2}{3})$
CW P32 (W4.4)	$D^2y = 12z$	$0, 0$	$2\log^3 x$
CW P33 (W4.5)	$(D^2 + 3D + 2)y = e^z + e^{-z}$	$-1, -2$	$\frac{x}{6} + \frac{\log x}{x}$
Jan s3 (W4.6)	$(D-2)(D-3)y = e^{4z}\sin e^z$	$2, 3$	$-x^2\sin x$

S011L·8 · CW U2 P29 · CW U2 P30 · CW U2 P31 · CW U2 P32 · CW U2 P33 · ESA Dec-23 3b · ESA Jul-23 3a · ESA Jan-24 s1 3c · ESA Jan-24 s2 3c · ESA Jun-24 3b · ESA Jan-24 s3 3d · T011Lb 00:04:59

Printed answers that are wrong

Source	Printed	Correct
Slides 011L·4–5	$k_2(\cdot)^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$	$k_2(\cdot)^{n-2}\frac{d^{n-2}y}{dx^{n-2}}$
CW P29	$-\log(1+x) - \cos\log(1+x)$	$-\log(1+x)\cos\log(1+x)$
CW P30	$C_2[(2+3x)^2]$	$C_2(2+3x)^{-2}$
CW P31	$-\frac{x^3}{3}(\log x + \frac{2}{3})$	$-\frac{x^2}{3}(\log x + \frac{2}{3})$
CW P32	$-2(\log x)^2$	$+2(\log x)^3$

CW U2 P29 · CW U2 P30 · CW U2 P31 · CW U2 P32 · S011L·4 · S011L·5

Important points and why

1. Polynomial right-hand sides give e^{kz} terms plus a constant: ESA Jan-24 s1 3c ($6x+5 \to 3e^z + 2$), s2 3c ($3x+4 \to 3e^z - 2$), Jun-24 3b ($8x^2 - 2x + 3 \to 2e^{2z} + 3e^z + 4$), CW P30 ($\to \frac{e^{2z}-1}{3}$).
   ESA ★★★ ESA Jan-24 s1 3c · ESA Jan-24 s2 3c · ESA Jun-24 3b · CW U2 P30 · S011L·8
2. Check every e^{kz} against the AE roots: the failure case appears in Jul-23 3a ($(D-2)^2$, twice), Jan-24 s2 3c ($(D-1)^2$, twice), Jun-24 3b (e^z, root 1), CW P29 ($\sin z$, roots $\pm i$) and CW P30 (e^{2z}, root 2).
   ESA ★★★ ESA Jul-23 3a · ESA Jan-24 s2 3c · ESA Jun-24 3b · CW U2 P29 · CW U2 P30
3. A repeated root m gives $(ax+b)^m[c_1 + c_2\log(ax+b)]$; complex roots $\pm i\beta$ give $c_1\cos(\beta\log(ax+b)) + c_2\sin(\beta\log(ax+b))$. ESA ★★★ ESA Jul-23 3a · CW U2 P29 · ESA Dec-23 3b

Why is a constant on the right handled as e^{0z}?
$c = ce^{0\cdot z}$, so Type 1 gives $\frac{c}{f(0)}$ when $f(0) \neq 0$ (slide 011L·10: $-\frac{9}{4}\cdot\frac{1}{-3}$).
S011L·10 · T011Lb 00:16:27

Exam trap. CW P29's printed answer "$-\log(1+x)-\cos\log(1+x)$" should be the product $-\log(1+x)\cos[\log(1+x)]$. CW U2 P29

Exam trap. CW P30's printed answer repeats $(2+3x)^2$; the second CF term is $(2+3x)^{-2}$. CW U2 P30

Exam trap. ESA Dec-23 3b ($\sin(2\log(1+x))$, no failure) and CW P29 ($2\sin\log(1+x)$, failure) look alike but differ. ESA Dec-23 3b · CW U2 P29

Worked examples

W3.1 · $(2x+3)^2y''-2(2x+3)y'-12y=6x$ (slides 011L·8–10) ESA ★★★

GIVEN Solve $(2x+3)^2\frac{d^2y}{dx^2}-2(2x+3)\frac{dy}{dx}-12y=6x$.

STEP 1 Legendre with $a=2$, $b=3$: $z=\log(2x+3)$, $2x+3=e^z$, $x=\frac{e^z-3}{2}$; $(2x+3)y'=2Dy$, $(2x+3)^2y''=4D(D-1)y$.

STEP 2 $(4D(D-1)-2\cdot 2D-12)y=6\cdot\frac{e^z-3}{2}$, i.e. $(4D^2-8D-12)y=3(e^z-3)$, so $(D^2-2D-3)y=\frac{3}{4}(e^z-3)=\frac{3e^z}{4}-\frac{9}{4}$.

STEP 3 CF: AE $m^2-2m-3=0$, $m=3,-1$: $y_c=c_1e^{3z}+c_2e^{-z}=c_1(2x+3)^3+\frac{c_2}{2x+3}$.

STEP 4 PI: $\frac{3e^z}{4(D^2-2D-3)}-\frac{9e^{0z}}{4(D^2-2D-3)}=\frac{3e^z}{4(-4)}-\frac{9}{4(-3)}=-\frac{3e^z}{16}+\frac{3}{4}$.

STEP 5 $y_p=\frac{3}{4}-\frac{3(2x+3)}{16}\ \left(=\frac{3}{16}-\frac{3x}{8}\right)$.

ANSWER $$\boxed{y=c_1(2x+3)^3+\frac{c_2}{2x+3}+\frac{3}{4}-\frac{3(2x+3)}{16}}$$

CHECK Slides 011L·8–10 agree (they stop at "$y=y_c+y_p$"); T011Lb agrees. sympy checkodesol: True. Figure G3.1, Figure F3.1.

S011L·8 · S011L·9 · S011L·10 · T011Lb 00:12:26 · T011Lb 00:17:39

W3.2 · $(1+x)^2y''+(1+x)y'+y=2\sin[\log(1+x)]$ (CW P29) ESA ★★

GIVEN Solve $(1+x)^2\frac{d^2y}{dx^2}+(1+x)\frac{dy}{dx}+y=2\sin[\log(1+x)]$.

STEP 1 $a=1$, $b=1$: $z=\log(1+x)$: $(D(D-1)+D+1)y=2\sin z$, i.e. $(D^2+1)y=2\sin z$.

STEP 2 AE $m^2+1=0$, $m=\pm i$: $y_c=c_1\cos z+c_2\sin z$.

STEP 3 PI: $D^2=-1$ makes the denominator 0 (failure): $\frac{2\sin z}{D^2+1}=2\cdot\left(-\frac{z}{2}\cos z\right)=-z\cos z$.

STEP 4 Back to x: $z=\log(1+x)$.

ANSWER $$\boxed{y=C_1\cos[\log(1+x)]+C_2\sin[\log(1+x)]-\log(1+x)\cos[\log(1+x)]}$$

CHECK sympy checkodesol: True. **CW prints "$-\log(1+x)-\cos\log(1+x)$"** (a sum, and "lox"): the sum leaves residual $-\log(1+x)-2\sin\log(1+x)$; the product is right. Figure G2.2.

CW U2 P29

W3.3 · $(1+x)^2y''+(1+x)y'+y=\sin(2\log(1+x))$ (ESA Dec-23 3b) ESA ★★★

GIVEN Solve $(1+x)^2\frac{d^2y}{dx^2}+(1+x)\frac{dy}{dx}+y=\sin 2(\log(1+x))$ (6 marks), read as $\sin(2\log(1+x))$.

STEP 1 $z=\log(1+x)$: $(D^2+1)y=\sin 2z$ (same left side as W3.2).

STEP 2 $y_c=c_1\cos z+c_2\sin z$.

STEP 3 PI: $D^2=-4$: $\frac{\sin 2z}{-4+1}=-\frac{1}{3}\sin 2z$ (no failure: $\pm 2i$ are not roots).

STEP 4 If $\sin^2(\log(1+x))$ were meant: $\sin^2 z=\frac{1-\cos 2z}{2}$, PI $=\frac{1}{2}+\frac{1}{6}\cos 2z$.

ANSWER $$\boxed{y=c_1\cos[\log(1+x)]+c_2\sin[\log(1+x)]-\frac{1}{3}\sin[2\log(1+x)]}$$

CHECK sympy: residual 0 for both readings. Compare CW P29 ($2\sin z$: failure case, W3.2).

ESA Dec-23 3b

W3.4 · $(2+3x)^2y''+3(2+3x)y'-36y=3x^2+4x+1$ (CW P30) ESA ★★

GIVEN Solve $(2+3x)^2\frac{d^2y}{dx^2}+3(2+3x)\frac{dy}{dx}-36y=3x^2+4x+1$.

STEP 1 $a=3$, $b=2$: $z=\log(3x+2)$, $x=\frac{e^z-2}{3}$; $(9D(D-1)+9D-36)y=9(D^2-4)y$.

STEP 2 RHS: $3x^2+4x+1=\frac{(e^z-2)^2}{3}+\frac{4(e^z-2)}{3}+1=\frac{e^{2z}-1}{3}$. So $(D^2-4)y=\frac{e^{2z}-1}{27}$.

STEP 3 AE $m=\pm2$: $y_c=c_1e^{2z}+c_2e^{-2z}=c_1(3x+2)^2+c_2(3x+2)^{-2}$.

STEP 4 PI: $\frac{1}{27}\frac{e^{2z}}{D^2-4}$ fails ($m=2$ is a root): $\frac{1}{27}\cdot\frac{ze^{2z}}{2D}\big|_{D=2}=\frac{ze^{2z}}{108}$; $-\frac{1}{27}\cdot\frac{1}{-4}=\frac{1}{108}$.

ANSWER $$\boxed{y=C_1(3x+2)^2+C_2(3x+2)^{-2}+\frac{1}{108}\left[(3x+2)^2\log(3x+2)+1\right]}$$

CHECK sympy checkodesol: True. **CW prints** $C_2[(2+3x)^2]$ for the second CF term (a duplicate); it is $(2+3x)^{-2}$. The PI agrees.

CW U2 P30

W3.5 · $(2x+1)^2y''-6(2x+1)y'+16y=8(2x+1)^2$ (ESA Jul-23 3a = May-24 3a) ESA ★★★

GIVEN Solve $(2x+1)^2\frac{d^2y}{dx^2}-6(2x+1)\frac{dy}{dx}+16y=8(2x+1)^2$ (7 marks).

STEP 1 $z=\log(2x+1)$, $a=2$: $(4D(D-1)-12D+16)y=8e^{2z}$, i.e. $4(D^2-4D+4)y=8e^{2z}$, $(D-2)^2y=2e^{2z}$.

STEP 2 AE $(m-2)^2=0$, $m=2,2$: $y_c=(c_1+c_2z)e^{2z}$.

STEP 3 PI: double failure: $\frac{2e^{2z}}{(D-2)^2}=2\cdot\frac{z^2}{2}e^{2z}=z^2e^{2z}$.

STEP 4 Back: $e^{2z}=(2x+1)^2$, $z=\log(2x+1)$.

ANSWER $$\boxed{y=(2x+1)^2\left[c_1+c_2\log(2x+1)+\log^2(2x+1)\right]}$$

CHECK sympy checkodesol: True. Figure F3.2.

ESA Jul-23 3a · ESA May-24 3a

W3.6 · $(2x+1)^2y''-2(2x+1)y'-12y=6x+5$ (ESA Jan-24 s1 3c) ESA ★★★

GIVEN Solve $(2x+1)^2y''-2(2x+1)y'-12y=6x+5$ (6 marks).

STEP 1 $z=\log(2x+1)$, $x=\frac{e^z-1}{2}$: $(4D(D-1)-4D-12)y=3(e^z-1)+5$, i.e. $4(D^2-2D-3)y=3e^z+2$.

STEP 2 AE $m=3,-1$: $y_c=c_1(2x+1)^3+\frac{c_2}{2x+1}$.

STEP 3 PI: $\frac{1}{4}\left[\frac{3e^z}{1-2-3}+\frac{2}{-3}\right]=-\frac{3e^z}{16}-\frac{1}{6}$.

ANSWER $$\boxed{y=c_1(2x+1)^3+\frac{c_2}{2x+1}-\frac{3(2x+1)}{16}-\frac{1}{6}}$$

CHECK sympy checkodesol: True ($y_p=-\frac{3x}{8}-\frac{17}{48}$). Same operator as the slide example (W3.1).

ESA Jan-24 s1 3c

W3.7 · $(x+2)^2y''-(x+2)y'+y=3x+4$ (ESA Jan-24 s2 3c) ESA ★★★

GIVEN Solve $(x+2)^2y''-(x+2)y'+y=3x+4$ (6 marks).

STEP 1 $a=1$, $b=2$: $z=\log(x+2)$, $x=e^z-2$: $(D(D-1)-D+1)y=3(e^z-2)+4$, i.e. $(D-1)^2y=3e^z-2$.

STEP 2 AE $m=1,1$: $y_c=(c_1+c_2z)e^z=(x+2)[c_1+c_2\log(x+2)]$.

STEP 3 PI: $\frac{3e^z}{(D-1)^2}=\frac{3z^2}{2}e^z$ (double failure); $\frac{-2}{(0-1)^2}=-2$.

ANSWER $$\boxed{y=(x+2)\big[c_1+c_2\log(x+2)\big]+\frac{3}{2}(x+2)\log^2(x+2)-2}$$

CHECK sympy checkodesol: True.

ESA Jan-24 s2 3c

W3.8 · $(2x-1)^2y''+(2x-1)y'-2y=8x^2-2x+3$ (ESA Jun-24 3b) ESA ★★★

GIVEN Solve $(2x-1)^2\frac{d^2y}{dx^2}+(2x-1)\frac{dy}{dx}-2y=8x^2-2x+3$ (7 marks).

STEP 1 $a=2$, $b=-1$: $z=\log(2x-1)$, $x=\frac{e^z+1}{2}$: $(4D(D-1)+2D-2)y=(4D^2-2D-2)y$.

STEP 2 RHS: $8x^2=2(e^z+1)^2=2e^{2z}+4e^z+2$, $-2x=-e^z-1$: total $2e^{2z}+3e^z+4$.

STEP 3 AE $4m^2-2m-2=0$, $2(2m+1)(m-1)=0$, $m=1,-\frac{1}{2}$: $y_c=c_1(2x-1)+c_2(2x-1)^{-1/2}$.

STEP 4 PI: $\frac{2e^{2z}}{16-4-2}=\frac{e^{2z}}{5}$; $\frac{3e^z}{4D^2-2D-2}$ fails at $D=1$: $\frac{3ze^z}{8D-2}\big|_{D=1}=\frac{ze^z}{2}$; $\frac{4}{-2}=-2$.

ANSWER $$\boxed{y=c_1(2x-1)+\frac{c_2}{\sqrt{2x-1}}+\frac{(2x-1)^2}{5}+\frac{(2x-1)\log(2x-1)}{2}-2}$$

CHECK sympy checkodesol: True (valid for $x>\frac{1}{2}$).

ESA Jun-24 3b

Past-paper questions

ESA Dec 2023 · 6 marks · UE22MA141A Q3.b · solved in W3.3

Solve $(1+x)^2\frac{d^2y}{dx^2}+(1+x)\frac{dy}{dx}+y=\sin 2(\log(1+x))$.

ESA Jul 2023 · 7 marks · UE22MA141A Q3.a · solved in W3.5

Solve $(2x+1)^2\frac{d^2y}{dx^2}-6(2x+1)\frac{dy}{dx}+16y=8(2x+1)^2$.

ESA May 2024 · 7 marks · UE22MA141A Q3.a · solved in W3.5

Solve $(2x+1)^2\frac{d^2y}{dx^2}-6(2x+1)\frac{dy}{dx}+16y=8(2x+1)^2$ (repeat of Jul-23 3a).

ESA Jan 2024 set 1 · 6 marks · UE23MA141A Q3.c · solved in W3.6

Solve $(2x+1)^2y''-2(2x+1)y'-12y=6x+5$.

ESA Jan 2024 set 2 · 6 marks · UE23MA141A Q3.c · solved in W3.7

Solve $(x+2)^2y''-(x+2)y'+y=3x+4$.

ESA Jun 2024 · 7 marks · UE23MA141A Q3.b · solved in W3.8

Solve $(2x-1)^2\frac{d^2y}{dx^2}+(2x-1)\frac{dy}{dx}-2y=8x^2-2x+3$.

Short-note recap

Slide: $(2x+3)^2y''-2(2x+3)y'-12y=6x \to (D^2-2D-3)y=\frac{3}{4}(e^z-3)$, $y=c_1(2x+3)^3+\frac{c_2}{2x+3}+\frac{3}{4}-\frac{3(2x+3)}{16}$. ESA: Jul-23 $(D-2)^2y=2e^{2z}$, $y=(2x+1)^2[c_1+c_2\log(2x+1)+\log^2(2x+1)]$; Jan s2 $(D-1)^2y=3e^z-2$; Jun-24 $(4D^2-2D-2)y=2e^{2z}+3e^z+4$; Dec-23 $(D^2+1)y=\sin 2z$, $y_p=-\frac{1}{3}\sin(2\log(1+x))$.

4 Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d

With $x = e^z$: $x^m \to e^{mz}$, $\log x \to z$, $\frac{1}{x} \to e^{-z}$. The lecture solves a third-order equation with complex AE roots and a failure case, and turns $xy''' + 10y''' = 0$ into Cauchy form by multiplying by x^3. CW P32 must first be multiplied by x^2. The ESA question $x^2y'' - 4xy' + 6y = x^4 \sin x$ gives $e^{4z}\sin(e^z)$, a non-standard right-hand side: use $\frac{1}{D-m}X = e^{mz}\int Xe^{-mz}dz$ (class 009) or variation of parameters in x; the PI is $-x^2 \sin x$.

Formulas

E4.1 · Cauchy: right-hand side in z ESA ★★★

$$x = e^z: \quad x^m \to e^{mz}, \qquad \frac{1}{x} \to e^{-z}, \qquad \log x \to z, \qquad x^m \log x \to z\, e^{mz}$$

GIVES The PI type.

APPLIES Every Cauchy problem.

NOTE $x^4 \sin x \to e^{4z}\sin(e^z)$ is not a standard form (E4.2).

T011Lb 00:06:06 · T011Lb 00:06:23 · CW U2 P31 · ESA Jan-24 s3 3d

E4.2 · Non-standard right-hand side: one factor at a time (class 009) ESA ★★

$$\frac{1}{D-m}X = e^{mz}\int X\, e^{-mz}\, dz$$

GIVES The PI of $(D-m_1)(D-m_2)y = X$ in two integrations.

APPLIES ESA Jan-24 s3 3d ($e^{4z}\sin e^z$).

NOTE Or divide by x^2 and use variation of parameters in x (class 011 VoP).

ADDED FOR CLARITY ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

Graphs and diagrams

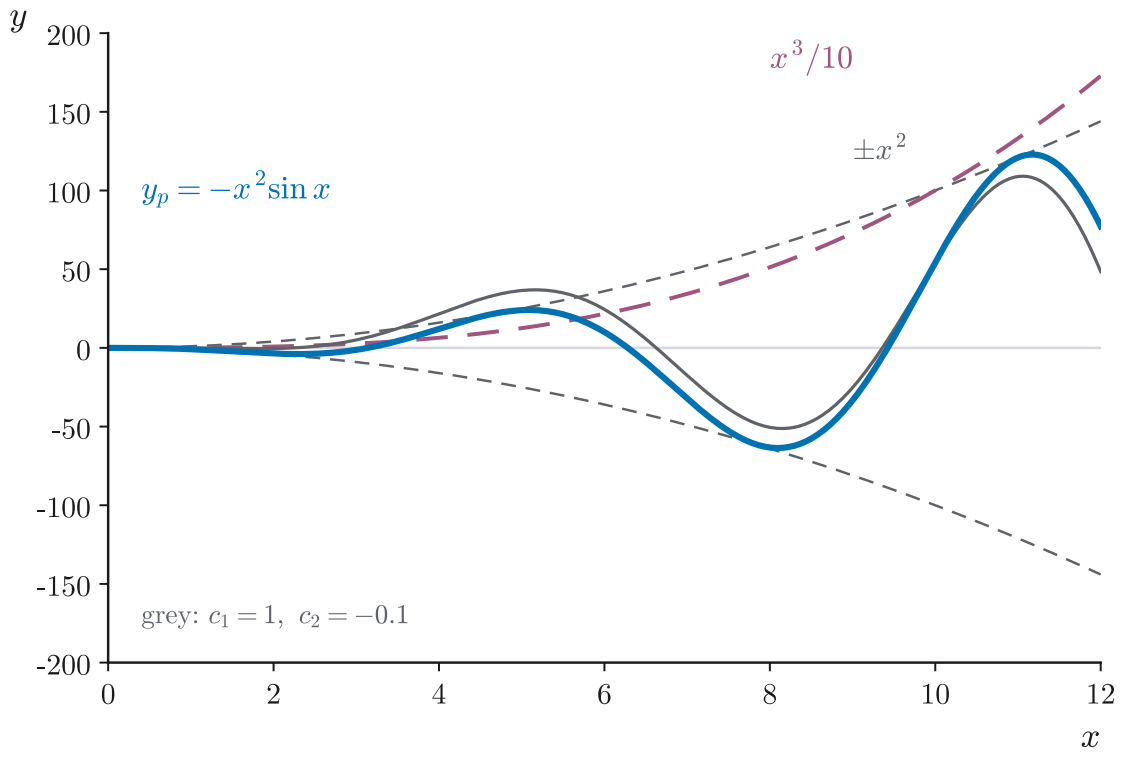
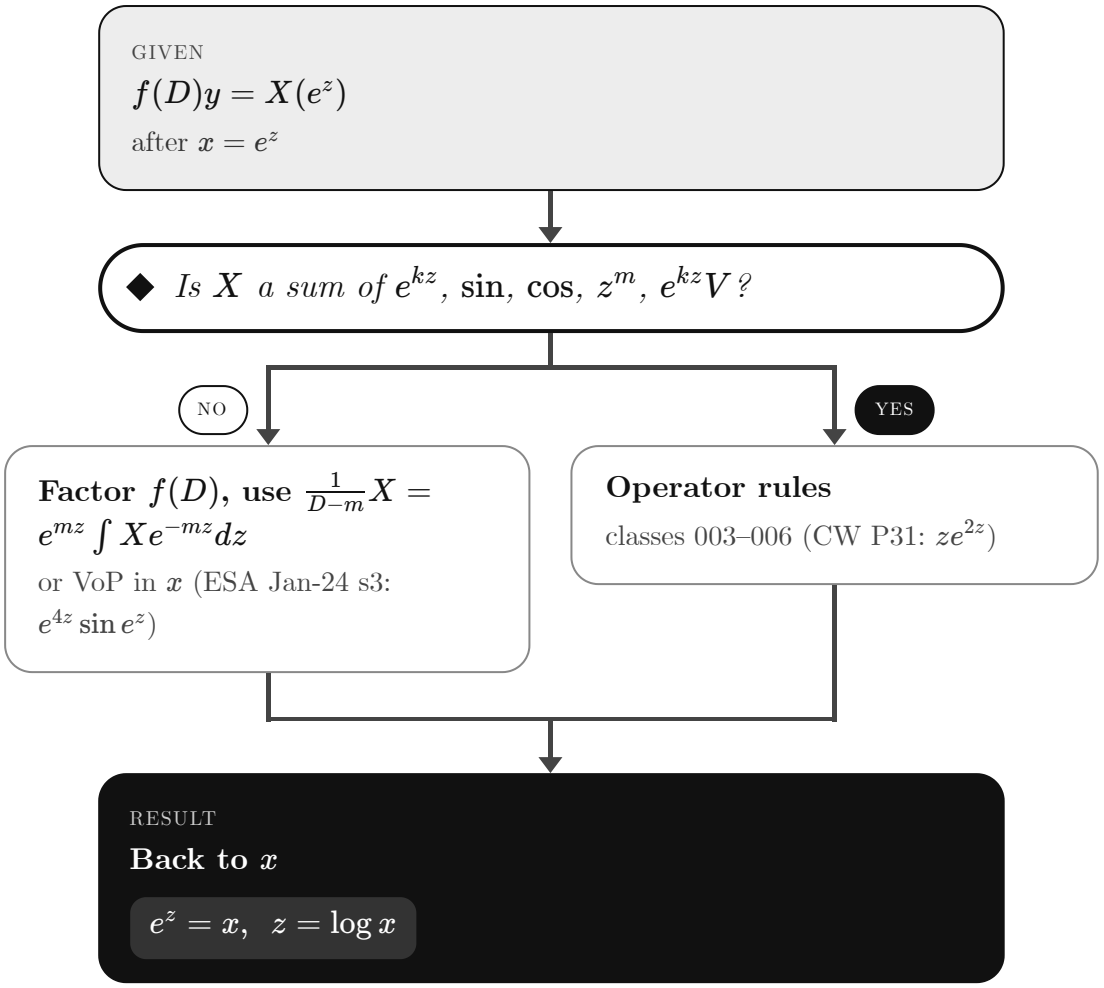


Figure G4.1. ESA Jan-24 s3 3d: $y_p = -x^2 \sin x$ and the CF terms x^2, x^3. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s3 3d

How to read it. For $x > 0$. Blue: $y_p = -x^2 \sin x$, oscillating between the grey envelopes $\pm x^2$. The upper envelope is the CF term x^2 itself; purple dashed: the CF term $\frac{x^3}{10}$ (scaled). Grey solid: the general solution with $c_1 = 1$, $c_2 = -0.1$.

Flowcharts



Remark. ESA Jan-24 s3 3d: $y = c_1x^2 + c_2x^3 - x^2\sin x$ (W4.6).

Figure F4.1. Is the right-hand side in z a standard form? ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s3 3d · CW U2 P31 · CW U2 P33

Important points and why

1. $x = e^z$: $x^m \to e^{mz}$, $\frac{1}{x} \to e^{-z}$, $\log x \to z$; so $x^2\log x$ is ze^{2z} (Type 4, class 006). ESA ★★★ T011Lb 00:06:23 · CW U2 P31 · CW U2 P33
2. Roots -1, $1 \pm i$ in z give $\frac{c_1}{x} + x[c_2\cos(\log x) + c_3\sin(\log x)]$. ESA ★★ TRANSCRIPT ONLY T011Lb 00:08:47 · T011Lb 00:09:11
3. ESA Jan-24 s3 3d: $e^{4z}\sin(e^z)$ is not a standard form; factor $(D-2)(D-3)$ and integrate twice, or use variation of parameters in x; $y = c_1x^2 + c_2x^3 - x^2\sin x$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

Exam trap. CW P31 prints $-\frac{x^3}{3}(\log x + \frac{2}{3})$ and CW P32 prints $-2(\log x)^2$; the correct PIs are $-\frac{x^2}{3}(\log x + \frac{2}{3})$ and $2(\log x)^3$. CW U2 P31 · CW U2 P32

Exam trap. CW P32 is not in Cauchy form as printed: multiply by x^2 first ($x^2y'' + xy' = 12\log x$). CW U2 P32

Worked examples

W4.1 · $x^3y''' + 2x^2y'' + 2y = 10\left(x + \frac{1}{x}\right)$ (lecture) ESA ★★

GIVEN Solve $x^3\frac{d^3y}{dx^3} + 2x^2\frac{d^2y}{dx^2} + 2y = 10\left(x + \frac{1}{x}\right)$ (Cauchy, order 3).

STEP 1 $x = e^z$: $[D(D-1)(D-2) + 2D(D-1) + 2]y = 10(e^z + e^{-z})$, i.e. $(D^3 - D^2 + 2)y = 10(e^z + e^{-z})$.

STEP 2 AE $m^3 - m^2 + 2 = 0$: $m = -1$ by inspection, then $m^2 - 2m + 2 = 0$, $m = 1 \pm i$. $y_c = c_1e^{-z} + e^z(c_2\cos z + c_3\sin z)$.

STEP 3 PI: $\frac{10e^z}{1-1+2} = 5e^z$; $\frac{10e^{-z}}{D^3-D^2+2}$ fails at -1: $\frac{10ze^{-z}}{3D^2-2D}\big|_{D=-1} = \frac{10ze^{-z}}{5} = 2ze^{-z}$.

STEP 4 Back: $e^z = x$, $z = \log x$.

ANSWER $$\boxed{y = \frac{c_1}{x} + x\left[c_2\cos(\log x) + c_3\sin(\log x)\right] + 5x + \frac{2\log x}{x}}$$

CHECK sympy checkodesol: True. T011Lb 00:11:38 is garbled (“5 into 10 by 52”); 00:11:56 states $5x + \frac{2}{x}\log x$.

TRANSCRIPT ONLY T011Lb 00:04:59 · T011Lb 00:06:46 · T011Lb 00:08:47 · T011Lb 00:11:56

W4.2 · $xy'''' + 10y''' = 0$ (lecture; multiply by x^3) ESA ★

GIVEN Find the general solution of $x\frac{d^4y}{dx^4} + 10\frac{d^3y}{dx^3} = 0$.

STEP 1 Not yet Cauchy (degree ≠ order). Multiply by x^3: $x^4y'''' + 10x^3y''' = 0$.

STEP 2 $x = e^z$: $D(D-1)(D-2)(D-3)y + 10D(D-1)(D-2)y = 0$, i.e. $D(D-1)(D-2)(D+7)y = 0$ (take the common factor).

STEP 3 AE roots $0, 1, 2, -7$ (real, distinct); homogeneous, so no PI: $y = c_1 + c_2e^z + c_3e^{2z} + c_4e^{-7z}$.

ANSWER $$\boxed{y = c_1 + c_2x + c_3x^2 + c_4x^{-7}}$$

CHECK sympy: residual 0. Direct check: $w = y'''$ satisfies $xw' + 10w = 0$, $w \propto x^{-10}$, so $y \ni x^{-7}$.

TRANSCRIPT ONLY T011Lb 00:17:56 · T011Lb 00:18:37 · T011Lb 00:21:08 · T011Lb 00:22:20

W4.3 · $x^2y'' - xy' - 3y = x^2\log x$ (CW P31) ESA ★★

GIVEN Solve $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = x^2\log x$.

STEP 1 $x = e^z$: $(D(D-1) - D - 3)y = ze^{2z}$, i.e. $(D^2 - 2D - 3)y = ze^{2z}$. AE $m = 3, -1$: $y_c = c_1x^3 + c_2x^{-1}$.

STEP 2 PI (Type 4, shift $D \to D+2$): $e^{2z}\frac{1}{(D+2)^2 - 2(D+2) - 3}z = e^{2z}\frac{1}{D^2+2D-3}z$.

STEP 3 $\frac{1}{-3(1-\frac{2D+D^2}{3})}z = -\frac{1}{3}\left(1 + \frac{2D}{3}\right)z = -\frac{1}{3}\left(z + \frac{2}{3}\right)$.

ANSWER $$\boxed{y = C_1x^{-1} + C_2x^3 - \frac{x^2}{3}\left(\log x + \frac{2}{3}\right)}$$

CHECK sympy dsolve: $C_1/x + C_2x^3 - \frac{x^2(3\log x + 2)}{9}$, the same. **CW prints** $-\frac{x^3}{3}(\log x + \frac{2}{3})$, which does not satisfy the equation.

CW U2 P31

W4.4 · $y'' + \frac{1}{x}y' = \frac{12\log x}{x^2}$ (CW P32) ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} = 12\frac{\log x}{x^2}$.

STEP 1 Multiply by x^2: $x^2y'' + xy' = 12\log x$ (Cauchy).

STEP 2 $x = e^z$: $(D(D-1) + D)y = 12z$, i.e. $D^2y = 12z$. AE $m = 0, 0$: $y_c = c_1 + c_2z$.

STEP 3 PI: $\frac{1}{D^2}12z$ = integrate twice: $6z^2$, then $2z^3$.

ANSWER $$\boxed{y = C_1\log x + C_2 + 2(\log x)^3}$$

CHECK sympy dsolve: $C_1 + C_2\log x + 2\log^3 x$. **CW prints** $-2(\log x)^2$: residual $-\frac{4(3\log x + 1)}{x^2} \neq 0$.

CW U2 P32

W4.5 · $x^2y'' + 4xy' + 2y = x + \frac{1}{x}$ (CW P33, homework) ESA ★★

GIVEN Solve $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = x + \frac{1}{x}$.

STEP 1 $x = e^z$: $(D(D-1) + 4D + 2)y = (D^2 + 3D + 2)y = e^z + e^{-z}$. AE $m = -1, -2$: $y_c = c_1x^{-1} + c_2x^{-2}$.

STEP 2 PI: $\frac{e^z}{1+3+2} = \frac{e^z}{6}$; e^{-z} fails ($m = -1$): $\frac{ze^{-z}}{2D+3}\Big|_{D=-1} = ze^{-z}$.

ANSWER $$\boxed{y = C_1x^{-1} + C_2x^{-2} + \frac{x}{6} + \frac{1}{x}\log x}$$

CHECK CW answer agrees; sympy checkodesol: True.

CW U2 P33

W4.6 · $x^2y'' - 4xy' + 6y = x^4 \sin x$ (ESA Jan-24 s3 3d = s4 3c) ESA ★★★

GIVEN Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^4 \sin x$ (6 marks).

STEP 1 $x = e^z$: $(D(D-1) - 4D + 6)y = (D-2)(D-3)y = e^{4z}\sin(e^z)$. AE $m = 2, 3$: $y_c = c_1x^2 + c_2x^3$.

STEP 2 PI, one factor at a time (E4.2): $\frac{1}{D-3}e^{4z}\sin e^z = e^{3z}\int e^z \sin(e^z)dz = -e^{3z}\cos(e^z)$ ($t = e^z$).

STEP 3 $\frac{1}{D-2}\big(-e^{3z}\cos e^z\big) = e^{2z}\int -e^z\cos(e^z)dz = -e^{2z}\sin(e^z) = -x^2\sin x$.

STEP 4 Check by variation of parameters in x: standard form $X = x^2\sin x$, $y_1 = x^2$, $y_2 = x^3$, $W = x^4$; $u = -\int x\sin x\,dx = x\cos x - \sin x$, $v = \int \sin x\,dx = -\cos x$; $uy_1 + vy_2 = -x^2\sin x$.

ANSWER $\boxed{y = c_1x^2 + c_2x^3 - x^2\sin x}$

CHECK sympy dsolve: $x^2(C_1 + C_2x - \sin x)$; both routes give $-x^2\sin x$. Figure G4.1, Figure F4.1.

ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

W4.7 · $x^2y'' - 3xy' + 3y = 12x^4$ by the operator rule (cf. class 011 VoP) ESA ★★

GIVEN The VoP lecture solves $x^2y'' - 3xy' + 3y = 12x^4$ by variation of parameters (VoP pack, W2.4). Solve it by the rules of this class.

STEP 1 $x = e^z$: $(D(D-1) - 3D + 3)y = (D^2 - 4D + 3)y = 12e^{4z}$. AE $m = 3, 1$: $y_c = c_1x^3 + c_2x$.

STEP 2 PI: $\frac{12e^{4z}}{16-16+3} = 4e^{4z} = 4x^4$.

ANSWER $\boxed{y = c_1x^3 + c_2x + 4x^4}$

CHECK sympy checkodesol: True; the same as by VoP, in one line.

ADDED FOR CLARITY T011Lb 00:04:06

Past-paper questions

ESA Jan 2024 set 3 · 6 marks · UE23MA141A Q3.d · solved in W4.6

Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^4\sin x$.

ESA Jan 2024 set 4 · 6 marks · UE23MA141A Q3.c · solved in W4.6

Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^4\sin x$ (repeat of set 3 Q3.d).

SHORT-NOTE RECAP

$x = e^z$: $x^m \to e^{mz}$, $\log x \to z$. CW P31 $y_p = -\frac{x^2}{3}(\log x + \frac{2}{3})$; P32 ($\times x^2$) $y = C_1 \log x + C_2 + 2\log^3 x$; P33 $y_p = \frac{x}{6} + \frac{\log x}{x}$. ESA $x^2y'' - 4xy' + 6y = x^4 \sin x$: $(D-2)(D-3)y = e^{4z}\sin e^z$, $y = c_1x^2 + c_2x^3 - x^2\sin x$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
a, b	Constants of the linear factor	$ax+b$ in Legendre's equation; Cauchy: $a=1$, $b=0$	E1.1, E2.1
$k_0, \ldots, k_n$	Coefficients	Constants multiplying each term; $k_0 \neq 0$	E1.1, E1.2, E2.5
X	Right-hand side	Function of x (capital); rewritten in z after the substitution (slide: $f(z)$)	E1.1, E2.5
z	New variable	$z=\log(ax+b)$ or $z=\log x$; independent variable after the substitution	E2.1
D	Operator	$\frac{d}{dz}$ after the substitution (clash: $\frac{d}{dx}$ in classes 002–009); some books θ	E2.2–E2.4
n	Order	Highest derivative; also the power of $(ax+b)$ on it	E1.1, E2.4
m	AE root	Root of the auxiliary equation in z; $e^{mz}=(ax+b)^m$	E2.6
k	Exponent of the RHS	e^{kz} terms of the right-hand side (clash with the coefficients k_i)	E3.1, E3.2
y_c, y_p	CF and PI	Complementary function and particular integral; $y=y_c+y_p$	W3.1
$\log$	Natural logarithm	$\log_e$; needs $ax+b>0$ ($x>0$ for Cauchy)	E2.1, E2.6

B Sources, coverage and gaps

Source	Path	Use
Deck 011 Legendre (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.*	11 slides; content on 4–10 (every equation read from the rendered images; .slides.md has no equations)
Transcript T011L (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.*	21.9 min; theory: both forms, the derivation of $(ax+b)^2y'' = a^2D(D-1)y$, Cauchy's substitution $x = e^z$
Transcript T011Lb (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.*	23 min; recap and three problems: Cauchy order 3, the slide example, $xy'''' + 10y''' = 0$
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 29–33 ("Problems on Legendre's linear differential equations"), page 6 rendered at 150 dpi; every answer recomputed
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Dec-23 3b, Jul-23 3a, May-24 3a, Jan-24 s1 3c, s2 3c, s3 3d, s4 3c, Jun-24 3b, via the shared ESA map (Dec-23 re-read)
Video 011 Legendre (.mp4, both parts)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation*.mp4	Not opened: every slide equation is readable, no transcript segment has avg_logprob below −0.5, and the one garbled number (T011Lb 00:11:38) is settled by the working; no video trigger applies

Eight ESA papers checked: UE22MA141A Dec 2023, Jul 2023, May 2024; UE23MA141A Jan 2024 sets 1–4, June 2024 (19 distinct Unit 2 questions, 24 occurrences). Legendre/Cauchy is the most-asked Unit 2 topic: 6 distinct, 8 occurrences, in all 8 papers, 6–7 marks each. Legendre: Dec-23 3b, Jul-23 3a = May-24 3a, Jan-24 s1 3c, s2 3c, Jun-24 3b; Cauchy: Jan-24 s3 3d = s4 3c. Substitution, operator results and RHS conversion ★★★; polynomial RHS (4×) and PI failure cases (4 of 6) ★★★; $\sin(k\log(ax+b))$ (1× + CW P29) ★★; general n-th order result and order-3/4 Cauchy (lecture only) ★★; multiplying $xy'''' + 10y''' = 0$ into Cauchy form ★.

Known gaps, conflicts and corrections

1. Slide 011L·5 prints the third term of the n-th order Legendre form as $k_2(ax+b)^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$, and slide 011L·4 prints the third term of Cauchy's form as $k_2x^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$: the derivative must be $\frac{d^{n-2}y}{dx^{n-2}}$ (degree of the coefficient = order of the derivative, as slide 011L·4's own Legendre line and T011L 00:04:06 say). Corrected in E1.1, E1.2 and a trap.

2. Slides 011L·8–10 (the worked example $(2x+3)^2y'' - 2(2x+3)y' - 12y = 6x$) were recomputed with sympy: the reduced equation $(D^2 - 2D - 3)y = \frac{3}{4}(e^z - 3)$, the CF $c_1(2x+3)^3 + \frac{c_2}{2x+3}$ and the PI $\frac{3}{4} - \frac{3(2x+3)}{16}$ are right. Slide 011L·10 ends with "Thus, $y = y_c + y_p$"; the general solution is written out in W3.1. Slide 011L·7 writes the reduced right-hand side as "$f(Z)$" (capital Z); it is the right-hand side X written in terms of z.

3. CW P29's answer is printed "$\cdots - \log(1+x) - \cos\log(1+x)$" (and "$\sin[\mathrm{lox}(1+x)]$"): the PI is the product $-\log(1+x)\cos[\log(1+x)]$; the printed sum does not satisfy the equation (sympy residual $-\log(1+x) - 2\sin\log(1+x)$). W3.2.

4. CW P30's printed answer "$C_1[(2+3x)^2] + C_2[(2+3x)^2] + \frac{1}{108}[(2+3x)^2\log(3x+2) + 1]$" repeats $(2+3x)^2$: the second CF term is $(2+3x)^{-2}$ (AE roots ± 2). The PI is right. W3.4.

5. CW P31's printed answer has $-\frac{x^3}{3}(\log x + \frac{2}{3})$; the PI is $-\frac{x^2}{3}(\log x + \frac{2}{3})$ (the right-hand side is $x^2\log x = ze^{2z}$, and 2 is not an AE root). The printed term does not satisfy the equation (sympy). W4.3.

6. CW P32's printed answer has $-2(\log x)^2$; the PI is $+2(\log x)^3$: in z the equation is $D^2y = 12z$, so $y_p = 2z^3$ (sympy dsolve gives $C_1 + C_2\log x + 2\log^3 x$). W4.4. CW P33's answer is right as printed.

7. ESA Dec-23 3b prints the right-hand side "$\sin 2(\log(1+x))$". It is read here as $\sin(2\log(1+x))$ (PI $-\frac{1}{3}\sin(2\log(1+x))$). CW P29 has the different right-hand side $2\sin[\log(1+x)]$ (a failure case, PI $-\log(1+x)\cos\log(1+x)$). If $\sin^2(\log(1+x))$ were meant, the PI would be $\frac{1}{2} + \frac{1}{6}\cos(2\log(1+x))$. All three are worked/checked in W3.2–W3.3.

8. ESA Jan-24 s3 3d ($x^2y'' - 4xy' + 6y = x^4\sin x$) becomes $(D^2 - 5D + 6)y = e^{4z}\sin(e^z)$ in z, which is not a standard PI form; it is solved by the class 009 rule $\frac{1}{D-m}X = e^{mz}\int Xe^{-mz}dz$ twice, and checked by variation of parameters in x (W4.6). Neither route is on the slides.

9. sympy's dsolve cannot solve the Legendre forms directly in x (NotImplementedError), so every Legendre answer was checked with checkodesol on the full general solution plus substitution at three points; the Cauchy answers were also checked with dsolve where it succeeds.

10. The deck shows no Cauchy substitution and no Cauchy example: $x = e^z$, $xy' = Dy$, $x^2y'' = D(D-1)y$, $x^3y''' = D(D-1)(D-2)y$ come from T011L 00:18:30–00:19:24 and T011Lb 00:04:06; the order-3 and order-4 examples from T011Lb (transcript only). Cauchy's equation is the CW sheet's "Self-learning component", yet it is asked in ESA Jan-24 s3 3d = s4 3c.

11. T011Lb 00:21:50–00:22:20 stops at "c_4x to the ...": the last term is c_4x^{-7} (root -7). T011Lb 00:11:38 reads the PI coefficient as "5 into 10 by 52"; the working gives $\frac{10}{5} = 2$, so $y_p = 5x + \frac{2\log x}{x}$ (as said at 00:11:56).

12. The general n-th order result $(ax+b)^n\frac{d^ny}{dx^n} = a^nD(D-1)\cdots(D-n+1)y$ is on the slides only for $n = 1, 2, 3$ (011L·6); the general form is added for clarity (sympy-checked on $(ax+b)^m$ for $n \le 3$).

13. Transcript recognition errors were corrected silently: "Legendary's", "legendary", "legend raise", "Letch and Grace", "Legendary C" → Legendre's; "Cautious differential equation", "causation d" → Cauchy's (equation); "scenear", "Lia differential" → linear differential; "n third or derivative", "n minus 1 third derivative" → n-th, $(n-1)$-th derivative; "delimitative", "derivativity", "therivative" → derivative; "dq", "mq" → D^3, m^3; "psi 1z", "c sign" → $\sin z$; "bz by dx" → $\frac{dz}{dx}$; "t square", "8 t" → D^2, $8D$; "of by" → of y; "P par Z", "epar z" → e^z; "e x plus three" → $2x+3$; "first return" → first term; "source 2" → is 2; "LD" → LDE.

Coverage

Source	Citations	Slides cited (slide × citations)
S011L	50	4×11, 5×10, 6×8, 7×4, 8×9, 9×4, 10×4
CW	35	
ESA	44	
T011L	50	
T011Lb	43	

The deck's own title slide says “SESSION: 9”; the bucket numbers it 011 and files both this deck and the variation-of-parameters deck (“SESSION: 8”) under 011 — hence the tags S011L and S011V (bucket numbers are not contiguous: the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11). Slides 1–3 and 11 are title, class-content and closing slides; slide 4 gives both forms (Legendre and Cauchy); slides 5–7 the substitution and operator results for Legendre; 8–10 one worked example. The deck has no Cauchy substitution and no Cauchy example: T011L (00:16:51–00:21:24) derives $x = e^z$, and T011Lb works two Cauchy examples (00:04:59, 00:17:56) besides the slide example (00:12:26). Cauchy's equation is the syllabus “self-learning component” (CW sheet header) but is asked in ESA Jan-24 s3/s4 and CW P31–P33, so it is covered fully here, flagged transcript only where it is not on the slides.